

DAYCLI_TOOLKIT: Software to Encode and Decoded Daily Climate Reports in WMO BUFR template 3-07-095 (Version 1.0)

1-Introduction and Historical Review

The development of the main measures of the state of the climate, such as the global temperature records, extensively depends on monthly CLIMAT data provided by National Meteorological and Hydrological Services (NMHSs). Over the last 30 years, there has been a growing demand for indices and measures of the climate that also consider means and extremes daily values. To address this issue, a new BUFR template for Daily Climate reports (template 3-07-074) was developed in 2016 and was named as DAYCLI message. see [1] and [2].

In 2018, a one-year trail phase of the use of DAYCLI message as launched by WMO. After this trail phase, a first revision of DAYCLI was done resulting in the development of a new DAYCLI message (template 3-07-075), which was included in the Manual on Code in 2020 [3]. A second revision was than done and it has been noted that the use of Coordinated Universal Time (UTC) in template 3 07 075 may introduce inconsistencies or potential misinterpretations for some National Meteorological and Hydrological Services (NMHSs), particularly those whose climatological observations are structured around their local time zones. Recognizing that most NMHSs measure, compute, and store daily climatological data according to national or regional time standards, **a new DAYCLI message (template 3-07-095)** was developed with collaboration of SERCOM / SC-CLI / ET-DDS, SERCOM / SC-CLI / ET-MCCVC and INFCOM / SC-IMT / ET-DATA STANDARD. It was concluded by ET-DATA in January 2026 [4] and will be included in The BUFR tables - version 46 during 2026. [5]

The software provided here, DAYCLI_TOOLKIT was developed at the Brazilian National Institute for Space Research (INPE) to support the encoding and decoding of DAYCLI BUFR messages under template **3-07-095**. In the present version there are two tools to encode DAYCLI BUFR message: The **daycli_encoder1** and **daycli_encoder2**. The **daycli_encoder1** can be use to encode observations which are measure at fixed and consistent times throughout the year. The **daycli_endoder2**can be use to encode data observation whose times may vary during the year. Another important tool is the **daycli_decoder**. It allows decoding a DAYCLI BUFR message back into the text format used by daycli_encode2, enabling the opening of DAYCLI BUFR files from other centers. It can also be used as a tool for verifying local encoding.

Notes:

- 1- This software does not perform climatological calculations or quality control decisions.*
- 2- The DAYCLI messages can be decoded by any available BUFR software, as well as by the daycli_decoder and/or bufr_dump tools included in this toolkit.*
- 3 - Simple examples is also included*

2- Requirements:

- A fortran 90 compiler or Higher. For example: gfortran, g95, ifort, pgf90, etc.
- Linux "make" command or Windows "nmake" command.

3 - Compilation in Linux

In case you already have gfortran installed, the daycli_toolkit can be compiled by just typing the command **make** in the terminal.

if you have another fortran compiler, check the file that contains the directives for different compilers in the directory :

- makefile_pgf90 (for pgf90 compiler),
- makefile_ifort (for intel fortran)
- makefile_g95 (for g95)

And edit the file "Makefile" to include the appropriate directive in your case, as in the example below

```
#-----  
# Include compilation directives for installed compiler  
#-----  
include makefile_gfortran  
#include makefile_g95  
#include makefile_ifort
```

Then type **make**

In case you are using another fortran compiler, that is not listed above, you can edit and include the appropriate directive in you case.

4 - Compilation in MS-Windows

The daycli_toolkit was developed in linux, but it can be compiled and used in Windows (DOS terminal or MS-Windows terminal).However, the g95 fortran compiler and nmake command from the Microsoft Visual Studio must be installed in your system.

Type the command below on DOS terminal or Windows terminal to compile

```
nmake Makefile_windows
```

5 - Environment variables

After compilation it is recommended that you set the environment **MBUFR_TABLES** in your system with the path to the folder where BUFR tables are.

The procedure for setting environment variables can be different in each system or environment. But they are simple and similar.

Here is an example for linux: In the case of **bufatables** saved in the path **/home/user/daycli_toolkit/bufrtables**, it is necessary to edit the **.bashrc** file (or equivalent) in your system and add the follow instruction

```
export MBUFR_TABLES=/home/user/daycli_toolkit/bufrtables
```

After the edition type on terminal the command **source ./bashrc** to apply changes.

6 - testing BUFR DAYCLI codification

6.1 - Running the test example

There are two examples of DAYCLI codification using daycli_toolkit: In the example/example_01 directory the script **run_example01.sh** demonstrates the encoding of daycli with **daycli_encoder1** , using text input file format-1; in the example/example_02, the script **run_example02.sh** demonstrates the encoding of daycli with **daycli_encoder2**, using text input file format-2.

Demostration of the decoding of daycli using two different tools: ****a) daycli_decoder**** (which decodes daycli back to the text) and ****b) bufr_dump**** are also included.

See and run the scripts **run_example01.sh** or **run_example02.sh** and see the results. See **item 7** for a general description of format 1 and format 2.

Notes this script is a bash script for linux. If you are using dos windows or windows terminal some adaptations can be necessary.

6.2 - Basic line command

To use this tool as a coding layer within the user's system, the basic command to be implemented follows one of the models below:

Using format1:

```
daycli_encoder1 -i "Name_of_input_text_file_format1.txt" -o "daycli_output_BUFR_file.bufr"
```

or using format2:

```
daycli_encoder2 -i "Name_of_input_text_file_format2.txt" -o "daycli_output_BUFR_file.bufr"
```

7 - Input data formats

There are two possible formats for the input data: The format used by daycli_encoder1 tool (format-1) and the format used by daycli_encoder2 (format-2). Both tools produce the exactly same BUFR DAYCLI message according to template **3-07-095**. The difference is that the daycli_encoder1 automatically calculates the time period of each variable, necessary for DAYCLI messages according to fixed parameters provided in format-1, while daycli_encoder2 does not process any calculations. It just encodes all information as it is provided in format-2. In both cases, the input data formats are based on "fortran namelist" format, in which groups of information are written in a structure that starts with "\$GroupName" and closes with "/" . More details about format-1 and 2 are provided below

7.1 Common groups in format 1 and format2

Both formats have the same initial groups as in the example

SECTION 1 group

```
&SECTION1  
CENTER= 85 !<-Identification of Originating/Generating  
SUBCENTER= 0 !<-Sub-center of generating center (See common code C12)  
/  

```

STATION_ID group

```
&STATION_ID  
LATITUDE= 46.45980  
LONGITUDE= -56.10750  
WIGOS=0-20000-0-71805  
WMO=71805  
HTEMP= 2 ! Height of temperature sensor  
HA= 21 ! Height of station above mean sea level  
SMC_TEMP=5! Siting and measurement quality classification for temperature  
SMC_PREC= ! Siting and measurement quality classification for precipitation  
METHOD_TM= 3 !Method used to calculate the average daily temperature  
TIME_OFFSET=-240! Time difference in minutes: Local Meteorological Time Zone -UTC  
! Use 0 in TIME_OFFICE in case of date and time in UTC  
/  

```

7.2 Input text Format 1

In addition to the groups presented in item 6.1, format-1 there are the following groups.

- The groups: &STIME_{TT,TN,TM,RR,DS} define a daily time period for each variable. For example for TN (Minimum temperature)

Minimum temperature

```
&STIME_TN  
HOUR=18
```

```
MINUTE=01
DT=-1
/
```

The example above indicates that minimum temperatures are measured from 18:01 LMTZ (HOUR=18, MINUTE=01) of the previous day (DT=-1) until 18:00 LMTZ of the current day (reference day). Use DT=0 to indicate that measurements are taken from 18:01 LMTZ of the reference day until 18:00 LMTZ of the following day.

With the information above the daycli_encoder-1 performs the calculation for time periods for all the minimum temperatures provided. For example. In case of DT=-1 and data assigned as minimum temperature of 1st March 2021 the software calculates the time period as 20210228T1801-20210301T1800 i e . from February 28, 2021 at 12:00 until March 1, 2021. In case of DT=1 the time period would be 20210301T1801-20210302T1800.

Note that the software considers the transitions between months or years to calculate the time period and then codes the information in DAYCLI BUFR message.

- The DATA_SECTION group

The last group, "&DATA_SECTION", is the group that contains the actual daily climate parameter on a table format with rows and columns separated by semicolons (";"). The columns in format 1 are: **date; rr; qrr; fsd; qfsd; tsd; qtsd; tn; qtn; tx; qtx; tm; qtm**

where

- **date** = Reference date LMTZ
- **rr** = Total Accumulate Precipitation (Kg m-2)
- **fsd** = Fresh snow depth (m)
- **tsd** = Total snow depth (m)
- **tx** = Maximum temperature (K)
- **tn** = Minimum temperature (K)
- **tm** = Mean temperature (K)

The quality flags for respective parameters above are identified by:

qrr, qfsd, qtsd, qtn, qtx, qtm

The example below shows the first few rows of the \$DATA_SECTION of the format1.

```
&DATA_SECTION
!date= Reference date and time in LMTZ / Attribution Day
!date ; rr; qrr; fsd; qfsd; tsd; qtsd; tn; qtn; tx; qtx; tm; qtm
20220401;15.5; 0; 0; 7; 0; 7;273.35; 0;278.95; 0; 275.05; 0
20220402;1.9;0;1;7;0;7;274.05;0;277.65;0;275.85;0
20220403;0.0;0;1;7;0;7;273.35;0;276.65;0;274.45;0
20220404;23.6;0;0;7;0;7;271.75;0;278.35;0;275.15;0
20220405;0.6;0;1;7;0;7;273.45;0;277.15;0;274.75;0
20220406;0.0;0;1;7;0;7;272.85;0;278.25;0;274.95;0
...
/
```

Complete examples of format 1 are included in the ./examples/example01/

7.3 Input text format 2

Format 2 does not have the &STIME_{TT,TN,TM,RR,DS} groups like format 1. Instead, the time periods are provided directly in the \$DATA_SECTION group. In other words, daycli_encoder2 does not perform any time window processing; it directly encodes the data in the DAYCLI message as it is provided in format 2.

In that way the \$DATA_SECTION group have the followed columns separated by semicolons (";").

**date; interval; tsd; qtsd; interval; rr; qrr; interval; fsd; qfsd; interval; tx
qtx;interval;tn;qtn;interval;tm;qtm**

where

- **date** = reference date in the format "yyyymmdd" (year, month and day)
- **interval (before tsd)** = Date and time of fresh snow depth measurement in format "yyyymmddThhmm" (Year, month, day, "T", hour and minute)
- **interval** = time interval used in processing extreme, average, or cumulative measurements that are provided in the following columns. The format is "yyyymmddThhmm-yyyymmddThhmm" to indicate the start and end time of the period
- **rr** = Total Accumulate Precipitation (Kg m-2)
- **fsd** = Fresh snow depth (m)
- **tsd** = Total snow depth (m)
- **tx** = Maximum temperature (K)
- **tn** = Minimum temperature (K)
- **tm** = Mean temperature (K)

The quality flag for each respective parameter above are identified by:

qrr, qfsd, qtsd, qtn, qtx, qtm

The example below shows the first few rows of the \$DATA_SECTION of the format2.

```
&DATA_SECTION
!date;interval;tsd;qtsd;interval;rr;qrr;interval;fsd;qfsd;interval;txT
qtx;interval;tn;qtn;interval;tm;qtm
20220401;20220401T0601;0;7;20220401T0601-20220402T0600;15.5;0;20220401T0601-
20220402T0600;0;7;20220401T0601-20220402T0600;278.95;0; 20200331T1801-
20220401T1800;273.35;0;20220401T0001-20220402T0000;275.05;0
20220402;20220402T0601;0;7;20220402T0601-20220403T0600;1.9;0;20220402T0601-
20220403T0600;1;7;20220402T0601-20220403T0600;277.65;0;20220401T1801-
20220402T1800;274.05;0;20220402T0001-20220403T0000;275.85;0
...
/
```

Complete examples of format 2 are included in the ./examples/example02/

8 - Notes

8.1 - The quality flag provide in format 1 and 2 are associated with respective parameter in codification process through the BUFR sequence 2-04-008; 0-31-021 which is set as 5. The correspondent table associated with "0-31-021" is "5" that can be found on "Code BUFR and Flag table" and it is also presented below:

- 0 = Data checked and declared good;
- 1 = Data checked and declared suspect;
- 2 = Data checked and declared aggregated;
- 3 = Data checked and declared out of instrument range;
- 4 = Data checked, declared aggregated, and out of instrument range;
- 5 = Parameter is not measured at the station;
- 6 = Daily value not provided;
- 7 = Data unchecked,
- 8-254 = Reserved;
- 255 = Missing (QC info not available)

8.2 - Software updates and other information can be obtained via the link https://github.com/sergioh-pessoal/daycli_toolkit or by contacting the author at sergioh.pessoal@gmail.com

9 - References

- [1] Jones, P.D., Lister, D.H., Osborn, T.J., Harpham, C., Salmon, M., Morice, C.P., 2012: Hemispheric and large-scale land-surface air temperature variations: An extensive revision and an update to 2010. *Journal of Geophysical Research*, 117, D05127, doi:10.1029/2011JD017139.
- [2] Propose of DAYCLI message at WMO/ET-DATA - (template 307075) (<https://github.com/wmo-im/BUFR4/issues/51>)
- [3] Manual on Codes (WMO-No 306), Volume 1.2 - FM94 BUFR edition 4
- [4] Propose of DAYCLI message at WMO/ET-DATA - (template 307095): "A new BUFR sequence for the exchange of daily summary report (DAYCLI)"(<https://github.com/wmo-im/BUFR4/issues/238>)
- [5] Latest Version of the Machine Readable Codes: Manual on Codes (WMO-No. 306), Volume I.2 Available on <https://wmo.int/latest-version>